Dilshad Sultana

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PROFESSIONAL SUMMARY: I am a recent graduate in Energy Science and Engineering from Khulna University of Engineering & Technology (KUET), with academic interests in renewable energy, energy systems, energy storage and sustainable engineering. My academic experience includes engineering simulation, research, technical projects and industrial exposure in power generation and energy technologies.

EDUCATION

B.Sc (Engineering) in Energy Science and Engineering **2022-2026**
Khulna University of Engineering and Technology, Khulna, Bangladesh
CGPA: 3.27 **Graduation: August 2026**
Average CGPA of the last **2 years (80 credits): 3.53 |** Best Term GPA**- 3.69**

Higher Secondary Certificate (HSC) in Science **2020**
Nawab Faizunnesa Govt. Girls High School, Cumilla, Bangladesh GPA: 5.00

Secondary School Certificate (SSC) in Science **2018**
Cumilla Shikkha Board Govt. Model College, Cumilla,Bangladesh GPA:5.00

COURSES

Energy & Thermal Engineering
- Thermodynamics
- Fluid Mechanics
- Heat and Mass Transfer
- Thermo-Fluid Devices

Fuels & Process Engineering
- Fuels and Engine Combustion
- Petroleum and Natural Gas Processing
- Hydrogen and Fuel Cells

Power & Energy Systems
- Electrical Circuits and Electronics
- Electrical Machines
- Power Electronics
- Power System Engineering
- Instrumentation and Control

Renewable & Sustainable Energy
- Solar Thermal Engineering
- Solar Photovoltaic Systems
- Bio and Wind Energy Engineering
- Energy Storage Systems
- Nuclear Power Engineering
- Power Plant Engineering

Building Energy & Management
- Energy Audit and Management
- HVAC&R System Design
- Energy in Built Environment

Simulation Courses
- Energy Engineering Simulation I (Ansys)
- Energy Engineering Simulation II (Design Builder)

SKILLS

TECHNICAL SKILLS:

Technical & Simulation Software: Python, MATLAB & Simulink, PCTran, Ansys

Engineering & Design Software: Solidworks, CAD, ASPEN HYSYS, Arduino EES, LogoSoft, DesignBuilder, Revit, Tinkercad

Productivity & Presentation: Microsoft PowerPoint, Word, Excel, Canva, Draw.io, LaTeX

SOFT SKILLS:
- Team Collaboration & Leadership
- Public Speaking & Technical Presentation
- Adaptability to New Tools and Technologies
- Analytical & Research Skills
- Project Management Skills

RESEARCH & PROJECTS

- Biodiesel Production from Blended Jojoba and Sesbania Oils: Optimization and Characterization **[Undergraduate Thesis]**
- Construction of a Portable Power Bank using **Solar Energy [UG Project]**
- Thermal Performance Assessment of Aerogel-Integrated Glazing System for Reduced Solar Heat Gain in Building using **Design Builder**
- Comparative analysis on the effect of break-size between the HOT and COLD legs during loss of AC power: A **PCTRAN**-based analysis
- Energetic and exergetic performance analysis of a Recuperative Organic Rankine Cycle modeled and validated using both **EES** and **Python**
- Fire Alarm System Integrated with Heat and Smoke Sensors (With LCD Display) in **TINKERCAD**
- Solar cell and panel simulation using **SIMULINK- MATLAB**

PUBLICATIONS

Conference Papers **Under Review**
- Production and Characterization of Biodiesel from Blended *Simmondsia chinensis* and *Sesbania cannabina* Oils
- Construction and Performance Evaluation of Solar-Powered Portable Power **[Submitted to ICMIEE 2026]**
- Comparative Energy Assessment of Single, Double and Aerogel Integrated Glazing for a Commercial Office Building in Dhaka using Design Builder **[Abstract Submitted to iCSNBE-2027]**

Journal Papers
Two manuscripts in preparation

RESEARCH INTEREST

Renewable Energy & Sustainability
Solar Photovoltaic Systems, Hybrid Renewable Energy Systems

Power & Electrical Systems
Smart Grid Technology, Microgrid Design and Control

INDUSTRIAL VISIT

Atomic Energy Research Establishment (Bangladesh)
Ganakbari, Savar, Dhaka-1349, Bangladesh

HAMKO and Abdullah Battery Company Limited
Jessore Road, Khulna-9100, Bangladesh

TRAINING

Data Analysis with Python and MATLAB **2024**
Under EDGE Training Program, ICT Division, Government of Bangladesh. Duration: 2months (80 hours)

Industrial Attachment **2024**
Khulna 225 MW Combined Cycle Power Plant under North-West Power Generation Company Ltd., Goalpara, Khalishpur, Khulna, Bangladesh. Duration: 7 days

Occupational Safety, Health and Environment Management 2024
Department of Energy Science and Engineering, KUET-9203, Khulna, Bangladesh Duration: 1 day

Training on International Environment Policy, Green Education, Leadership & Communication **2025**
Organized by Shishu Ullash Organization / International Global Youth Platform

EXPERIENCE

Practical business experience through family's electronic components business
Duration- **(2020- present)**

EXTRACURRICULAR ACTIVITIES

Club Activities
- Treasurer, Energy & Automation Club **2025-2026**
- Assistant Cultural Secretary
  Greater Cumilla Association of KUET (GCAK) **2023-2024**

Chittagong Divisional Debate Festival – 2016
Organized by NDF BD-VCDS & Mercantile Bank Ltd

SCHOLARSHIPS

University Merit Scholarship (3 times) 2022–23
Khulna University of Engineering & Technology (KUET)
- Achieved for consistently holding a top 50% merit position

Technical Scholarship
Cumilla Shikkha Board Govt. Model College, Cumilla, Bangladesh
- Earned for securing a top merit position in college

LANGUAGES

English (Fluent) | Bangla (Native)